

1 Cross-Load Generalization

To comprehensively evaluate the robustness of the proposed framework against varying operational loads, cross-load experiments were conducted on the SDUST dataset under two distinct configurations: single-factor domain shifts under a constant speed of 1000 rpm (Tasks RT1–RT3), and compounded dual-factor domain shifts featuring simultaneous speed fluctuations (1000–2000 rpm) and unseen loads (Tasks RT4–RT6). The detailed task configurations and experimental results are presented in Table 1 and Table 2, respectively.

Under the relatively simpler single-factor shifts (RT1–RT3), where distribution discrepancies are primarily driven by load-induced amplitude variations, most advanced domain generalization models exhibit strong performance. Both the proposed FDAN and the strongest baseline, IEDGNet, achieve near-perfect average accuracies exceeding 98.70%.

However, the true challenge emerges in Tasks RT4–RT6, where the compounding effects of load-induced stiffness variations and non-linear temporal warping caused by varying speeds introduce severe multi-source interferences. Under these dual-factor domain shifts, standard convolutional architectures are significantly impaired; for instance, the CNNC model experiences noticeable degradation, with its accuracy deteriorating to 78.20% on Task RT6. Furthermore, even highly competitive models like CDDG and IEDGNet experience performance drops and increased diagnostic variance. For example, IEDGNet’s accuracy drops from 99.20% on RT1 to 96.90% on RT4 when the speed shift is introduced.

In stark contrast, FDAN demonstrates exceptional resilience and stability. When confronted with these simultaneous speed fluctuations and unseen loads, FDAN sustains robust performance, achieving highest accuracies between 98.90% and 100.00% with exceptionally low standard deviations ($\leq 0.43\%$). This decisive superiority under compounded perturbations strongly validates the necessity of our framework’s design. By explicitly identifying task-relevant dimensions and employing targeted transport augmentation, FDAN successfully neutralizes multi-source interferences to extract strictly domain-invariant fault signatures, ensuring highly reliable diagnostics in complex, non-stationary industrial environments.

Table 1: Description of Cross-Load DG Tasks on the SDUST Dataset.

Task	Condition Load (N)	
	Source domains	Target domain
RT1	0, 40, 60	20
RT2	0, 20, 60	40
RT3	0, 20, 40	60
RT4	0, 40, 60	20
RT5	0, 20, 60	40
RT6	0, 20, 40	60

Table 2: Diagnostic Accuracy of Different Methods Under Cross-Load Scenarios (Tasks RT1 - RT6). (%)

Method	Task (Constant Speed: 1000 rpm)			Task (Variable Speed: 1000-2000 rpm)		
	RT1	RT2	RT3	RT4	RT5	RT6
CNNC	88.80 $\pm$ 2.48	89.00 $\pm$ 1.25	81.80 $\pm$ 1.82	82.20 $\pm$ 2.33	80.70 $\pm$ 2.79	78.20 $\pm$ 1.45
CCDG	91.80 $\pm$ 4.30	87.90 $\pm$ 6.78	90.90 $\pm$ 11.55	85.00 $\pm$ 4.73	85.90 $\pm$ 9.74	80.00 $\pm$ 7.44
CCN	91.70 $\pm$ 2.95	95.00 $\pm$ 5.24	88.00 $\pm$ 3.16	75.30 $\pm$ 4.00	79.20 $\pm$ 5.85	85.80 $\pm$ 5.57
DGNIS	79.40 $\pm$ 2.94	91.40 $\pm$ 1.67	84.50 $\pm$ 2.66	80.40 $\pm$ 4.94	81.80 $\pm$ 3.81	79.80 $\pm$ 2.16
DDD	97.30 $\pm$ 0.83	98.20 $\pm$ 1.25	98.20 $\pm$ 0.57	93.70 $\pm$ 1.20	97.60 $\pm$ 0.89	97.50 $\pm$ 0.70
CDDG	94.50 $\pm$ 2.64	98.40 $\pm$ 1.98	98.10 $\pm$ 1.38	92.60 $\pm$ 0.54	93.50 $\pm$ 1.27	95.20 $\pm$ 2.04
IEDGNet	99.20 $\pm$ 0.57	100.00 $\pm$ 0.00	99.90 $\pm$ 0.22	96.90 $\pm$ 1.57	98.40 $\pm$ 0.65	99.80 $\pm$ 0.27
FDAN	99.20 $\pm$ 0.20	100.00 $\pm$ 0.00	98.70 $\pm$ 0.33	98.90 $\pm$ 0.43	99.40 $\pm$ 0.30	100.00 $\pm$ 0.00

2 Cross-Sensor Spatial Generalization

To comprehensively evaluate the robustness of the proposed method against spatial variability and varying vibration transmission paths, cross-sensor domain generalization experiments were conducted. The dataset was provided by AECC Harbin Bearing Co., Ltd., comprising vibration signals collected under varying rotational speeds of 500, 1000, and 1500 r/min. The diagnostic task encompasses six health conditions: a normal state and inner race faults with five varying severity levels (0.6, 1.2, 2.1, 3.3, and 4.5 mm). To simulate real-world sensor placement variations, sensors A and B were mounted on the top surface of the bearing housing, while sensors D and E were arranged on the side surface. This spatial discrepancy inherently introduces significant domain shifts due to the completely different structural transmission paths of the vibration signals.

As summarized in Table 3, four cross-sensor Leave-One-Domain-Out tasks (RT7 to RT10) were formulated. While the baseline CNNC achieves respectable accuracies, it still exhibits vulnerability to structural transmission variations. In contrast, transferring knowledge across different sensors proves to be highly effective for FDAN, consistently achieving an average accuracy exceeding 98.80% across all tasks. The ability to maintain exceptional accuracy under spatial distribution shifts confirms that the framework successfully disentangles sensor-location-specific biases from the essential fault signatures. By overcoming acoustic and physical transmission variations, the proposed framework exhibits outstanding cross-sensor generalization capabilities, significantly enhancing its flexibility for deployment in complex industrial environments.

Table 3: Description and accuracy of cross-sensor domain generalization tasks.

Task	Sensor number		Accuracy	
	Source domains	Target domain	CNNC	FDAN
RT7	B, D, E	A	93.90 $\pm$ 1.19	98.80 $\pm$ 0.44
RT8	A, D, E	B	94.30 $\pm$ 1.02	98.90 $\pm$ 0.89
RT9	A, B, E	D	95.70 $\pm$ 1.35	99.20 $\pm$ 0.57
RT10	A, B, D	E	94.20 $\pm$ 1.52	99.60 $\pm$ 0.55

3 Extreme Cross-Hardware Generalization

To evaluate generalization capabilities under extreme domain shifts, a cross-hardware domain generalization experiment was conducted utilizing four diverse bearing fault datasets: MFPT, JNU, SEU, and XJTU. Based on empirical evidence that inner and outer race faults constitute the vast majority of bearing failures, the classification space was standardized into three fundamental categories: Normal, Inner Race Fault, and Outer Race Fault.

These four datasets exhibit profound heterogeneity. They were collected from entirely different mechanical test rigs, utilizing varying sensor types and sampling frequencies, and operating under disparate speeds and loads. To construct rigorous DG scenarios, Leave-One-Domain-Out tasks (RT11 to RT14) were designated, sequentially holding out one dataset as the unseen target domain.

Table 4: Description of extreme cross-dataset domain generalization tasks.

Task	Datasets		Accuracy	
	Source domains	Target domain	CNNC	FDAN
RT11	XJTU, JNU, SEU	MFPT	43.69 $\pm$ 5.53	57.48 $\pm$ 5.86
RT12	MFPT, JNU, SEU	XJTU	38.81 $\pm$ 5.59	46.94 $\pm$ 7.08
RT13	MFPT, XJTU, SEU	JNU	46.11 $\pm$ 5.12	53.89 $\pm$ 4.83
RT14	MFPT, XJTU, JNU	SEU	33.06 $\pm$ 2.24	43.69 $\pm$ 8.01

As presented in Table 4, cross-hardware generalization poses an immensely difficult challenge. Unlike cross-condition tasks within a single dataset, these tasks involve completely disjoint physical structures, different structural resonance frequencies, and distinct vibration transmission paths. Consequently, the baseline CNNC suffers severe performance collapse, dropping to accuracies as low as 33.06%. In contrast, FDAN demonstrates a pronounced advantage across all scenarios. For instance, on the highly volatile MFPT target domain (Task RT11),

which incorporates wildly varying load conditions ranging from 50 to 300 lbs entirely unrepresented in the source datasets, FDAN achieves an accuracy of 57.48%, outperforming the baseline by nearly 14%. Despite the extreme structural domain gaps that inherently restrict absolute performance, the consistent and significant margin of improvement over the baseline conclusively validates the framework’s superior ability to capture invariant core fault semantics under exceptionally challenging and disjoint physical conditions.

4 Robustness Against Extreme Industrial Noise and Interference

Experimental Details: To comprehensively evaluate the robustness of the proposed framework under extreme industrial noise and interference conditions, a rigorous evaluation was conducted on Tasks T6-T10. After the model was fully trained on pristine source domains, three types of typical industrial perturbations were artificially injected into the target domain data during the testing phase under three progressive severity levels. These perturbations were specifically selected to simulate common real-world deployability challenges:

- **Gaussian Noise:** Simulates ambient environmental noise and sensor electromagnetic interference. Mathematically, Additive White Gaussian Noise (AWGN) is injected by defining an amplitude Noise-to-Signal Ratio (NSR). Specifically, the severity level explicitly represents this NSR, formulated as the ratio of the noise standard deviation to the original signal standard deviation. For example, an NSR of 0.5 indicates that $\sigma_{\text{noise}} = 0.5\sigma_{\text{signal}}$.
- **Amplitude Scaling:** Simulates sudden operational load fluctuations or sensor calibration drift. The severity parameter defines the maximum boundary of amplitude fluctuation. For a given severity γ , the entire signal is multiplied by a random scalar drawn from a uniform distribution $\mathcal{U}(1 - \gamma, 1 + \gamma)$. Thus, a severity of 30% means the signal amplitude is randomly scaled anywhere between 70% and 130% of its original magnitude.
- **Signal Missing:** Simulates transient sensor disconnection or wireless transmission packet loss in Industrial IoT environments. The severity explicitly defines the temporal dropout ratio. A dropout ratio of 0.5 implies that 50% of the sampling points in the time-series sequence are randomly masked to zero, severely disrupting the temporal continuity and structural periodicity of the waveform.

As demonstrated in Table 5, the proposed framework exhibits remarkable resilience against complex industrial interferences.

Table 5: Diagnostic accuracy of the proposed method under various industrial perturbations on Tasks T6-T10.

Task	Gaussian Noise(NSR)			Amplitude Scaling(%)			Signal Missing			Original signal
	0.1	0.3	0.5	10	30	50	0.1	0.3	0.5	
T6	88.00	85.20	82.70	86.90	81.10	70.10	82.60	72.50	50.60	90.30
T7	91.30	89.40	80.90	89.10	78.40	69.30	87.40	74.60	44.10	93.20
T8	92.40	90.40	79.50	90.50	77.80	67.80	89.00	75.80	45.20	93.70
T9	91.80	87.50	81.80	90.40	80.00	69.20	85.80	75.10	48.50	93.80
T10	89.70	89.70	84.60	89.70	82.30	70.10	86.90	71.60	44.00	90.40

1) Resistance to Ambient Noise (Gaussian Noise): Resistance to Ambient Noise: Even under severe background noise with an NSR of 0.5, the model consistently maintains an accuracy above 79.50% across all tasks. For instance, on Task T8, the accuracy only experiences a marginal drop from 93.70% to 92.40% at an NSR of 0.1. This confirms that the proposed feature disentanglement mechanism effectively filters out high-frequency stochastic noise, retaining the core causal fault semantics.

2) Robustness to Load Fluctuations: When the signal amplitude fluctuates by up to 30%, the model sustains highly competitive accuracies ranging from 77.80% to 82.30%. This indicates that the framework does not merely memorize absolute vibration magnitudes, but successfully captures the structural and proportional relationships of the fault frequencies, ensuring stability against load-induced or speed-induced amplitude shifts.

3) Resilience to Sensor Dropout: Random signal missing presents the most extreme challenge, as it fundamentally destroys the temporal continuity of the waveform. Remarkably, at a dropout ratio of 0.1, the model retains strong diagnostic capabilities with accuracies exceeding 80%. While a dropout ratio of 0.5 inevitably causes a substantial

performance degradation to approximately 44% to 50% due to the severe corruption of structural periodicity, maintaining nearly half the accuracy under such catastrophic signal destruction still significantly outperforms random guessing.

Analysis: The framework exhibits remarkable resilience against complex real-world interferences. 1) Resistance to Ambient Noise: Even under severe background noise simulating sensor electromagnetic interference (NSR 0.5), FDAN consistently maintains an accuracy above 79.50%, confirming the disentanglement effectively filters high-frequency stochastic noise. 2) Robustness to Load Fluctuations: With sudden amplitude scaling up to 30%, accuracies remain highly competitive (77.80% to 82.30%), indicating the model captures the structural relationships of fault frequencies rather than memorizing absolute magnitudes. 3) Resilience to Sensor Dropout: Simulating severe Industrial IoT packet loss, random signal missing fundamentally destroys structural periodicity. Remarkably, at a 0.5 dropout ratio, maintaining nearly half the accuracy under such catastrophic signal destruction still significantly outperforms random guessing, showcasing the framework’s strict reliance on causal harmonics.

5 Hyperparameter Sensitivity Analysis (Batch Size & Imbalance)

1. Sensitivity to Batch Size: To evaluate the impact of batch size on the proposed framework, sensitivity experiments were conducted on Task T8, with the results summarized in Table R4.

It can be observed that at relatively small batch sizes, the model exhibits suboptimal diagnostic performance. Since the proposed method fundamentally relies on data-driven statistical estimations, a restricted batch size fails to provide a comprehensive representation of the global feature distribution. Consequently, the estimated class centroids lack statistical precision, which severely limits the directional diversity and effectiveness of the Task-relevant Transport Augmentation (TTA) module. As the batch size increases from 32 to 64, the accuracy rises steadily, peaking at 93.70%. This improvement stems from the fact that a larger batch encompasses sufficient intra-class samples to robustly model the underlying feature distribution, thereby enabling the TTA module to synthesize highly representative augmented features. However, when the batch size is further enlarged to 128, the performance plateaus at 93.60%. This indicates a point of diminishing returns: once the batch size is adequately large to establish reliable statistical estimations, further expanding it provides no additional diagnostic benefits.

Table R4: Accuracy of different batch sizes on Task T8.

Batch size	8	16	32	64	128
Accuracy	89.90 $\pm$ 0.93	90.40 $\pm$ 0.43	92.70 $\pm$ 0.20	93.70 $\pm$ 0.57	93.60 $\pm$ 0.67

2. Sensitivity to Extreme Class Imbalance: To evaluate the robustness of the proposed FDAN under severe industrial conditions—where specific fault data is notoriously difficult to acquire—we conducted targeted sensitivity experiments across varying degrees of induced class imbalance on the SDUST gearbox dataset (Task T13). The imbalance ratio (IMR) defines the probabilistic proportion of minority fault samples retained relative to the majority classes during training. In our specific experimental setup, classes representing a healthy state, planetary gear pitting, and planetary gear cracks were designated as the readily available majority classes, while planetary gear wear, sun gear pitting, sun gear cracks, and sun gear wear were treated as the scarce minority classes.

We first established an absolute baseline using a strictly balanced dataset, which yielded an accuracy of 97.30%. Under a near-balanced empirical setup (IMR = 1, simulating natural sampling variances), the model established a strong baseline accuracy of 97.40%. When the available minority class data is artificially halved (IMR = 0.5), the diagnostic performance remains remarkably stable, yielding 97.10% with a negligible degradation of merely 0.3%. Crucially, even under severe data scarcity, where the minority fault classes probabilistically retain only 10% of their original volume (IMR = 0.1), FDAN successfully sustains a highly competitive accuracy of 93.30%. While such extreme imbalance inevitably introduces a slight performance drop and an increased variance due to the profound lack of minority samples, maintaining an accuracy above 93% strongly validates the resilience of our framework. This robustness is primarily attributed to the synergistic effect of the Task-relevant Factor Identification (TFI) and the transport augmentation modules.

Instead of overfitting to the dominant prior probability distribution of the majority classes, the framework explicitly forces the network to isolate invariant causal features and dynamically synthesizes meaningful representations for the underrepresented minority classes, thereby effectively mitigating the risk of majority-class collapse.

Table R5: Accuracy across different Imbalance Ratios on Task T13.

Imbalance Ratio (IMR)	Balanced	1.0	0.5	0.1
Accuracy	97.30% $\pm$ 0.86%	97.40% $\pm$ 0.93%	97.10% $\pm$ 0.80%	93.30% $\pm$ 2.70%

6 Model Complexity and Inference Efficiency

To address your comment, we have conducted a comprehensive efficiency simulation to measure the total parameters, FLOPs, inference time per sample, and training time per batch for both our proposed FDAN and all comparison baselines. The efficiency simulation was evaluated on a single workstation equipped with an NVIDIA RTX 3090 GPU. The quantitative results are summarized in Table R7.

For the practical deployment of intelligent fault diagnosis systems, real-time inference capability and hardware resource consumption are as critical as diagnostic accuracy.

Table R7: Comparison of Model Complexity, Training Cost, Inference Efficiency, and Diagnostic Accuracy on the SDUST Dataset (Tasks T1 – T15). (%)

Model	Params (M)	FLOPs (M)	Inf. Time (ms/sample)	Train Cost (s/batch)	Accuracy (%)		
CNNC	0.162	19.058	0.0094	0.0082	52.62	75.53	77.26
CCN	0.161	19.058	0.0102	0.0371	53.35	77.93	81.08
CCDG	0.161	19.058	0.0100	0.0076	56.76	77.44	80.69
DGNIS	0.495	38.290	0.0226	0.1417	56.68	78.01	88.47
DDDG	1.201	19.058	0.0103	0.1983	57.57	87.75	89.94
CDDG	0.526	19.058	0.0101	0.0925	61.20	87.82	90.61
IEDGNet	0.219	19.058	0.0098	0.1392	61.17	88.54	93.79
FDAN	0.858	19.058	0.0098	0.1947	62.86	92.28	97.34

Note: Diagnostic accuracy results for Tasks T6–T15 are omitted.

1. Model Complexity (Parameters and FLOPs): As shown in Table R7, foundational baselines like CNNC, CCN, and CCDG require approximately 0.16 M parameters. In contrast, our proposed FDAN possesses 0.858 M parameters. This moderate increase is attributed to the explicit introduction of the Task-Relevant Factor Identification (TFI) module and the bidirectional feature transport architecture. Although FDAN has slightly more parameters than the lightest baselines, it remains exceptionally compact compared to heavier architectures like DDDG (1.201 M). More importantly, the inference computational complexity of FDAN is 19.058 M FLOPs, which is identical to most baseline methods. To guarantee experimental fairness, the base backbone architectures across all evaluated models share identical parameter settings. The reason FDAN introduces no extra inference overhead is that its complex auxiliary modules (such as the TFI and transport augmentation branches) are utilized exclusively during offline training. For actual real-time deployment, these components are discarded, leaving only the trained base Encoder and Classifier. Consequently, the inference FLOPs remain exactly consistent. This decisively demonstrates that our framework achieves rigorous feature disentanglement and augmentation during offline training without adding any computational burden or structural overhead to the forward pass during physical deployment.

2. Training Cost: The training cost of FDAN is 0.1947 seconds per batch. This is higher than the simpler alignment-based baselines but remains highly comparable to advanced domain generalization architectures like DDDG (0.1983 s/batch). This cost is a direct and necessary result of our staged, multi-objective optimization strategy and the gradient calculations required for sparse attention mask generation. Since model training in industrial applications is typically conducted offline on high-performance servers, this training duration is highly efficient and entirely acceptable for real-world development cycles.

3. Inference Efficiency: During inference, FDAN requires only 0.0098 ms per sample. This speed is virtually identical to the fastest baseline models (e.g., CNNC at 0.0094 ms and IEDGNet at 0.0098 ms) and is significantly faster than models like DGNIS (0.0226 ms). Processing a sample in under 0.01 milliseconds yields a theoretical throughput that vastly exceeds the real-time monitoring requirements of typical rotating machinery, ensuring zero-latency diagnostics at the industrial edge.

The slight increase in training cost and parameter count in FDAN is a deliberate and highly rewarding trade-off. Because the inference FLOPs and inference time remain identical to the most lightweight baselines, FDAN introduces zero additional delay during real-time deployment. In exchange, FDAN delivers substantial improvements in diagnostic accuracy under complex domain shifts (outperforming all baselines by notable margins, as shown in the accuracy columns). We conclude that FDAN strikes an optimal balance between computational efficiency and cross-domain robustness, making it highly viable for practical industrial deployment.

To explicitly address your concern regarding computational efficiency and practical deployability, we have incorporated a dedicated comparative analysis into the revised manuscript.

7 Quantitative Evaluation: Ablation Study on the Necessity of FFT:

To rigorously and empirically validate the indispensability of the FFT preprocessing step, we have conducted a comprehensive ablation study across both the baseline network (CNNC) and our proposed framework (FDAN). The diagnostic results across the DIGR dataset and the two distinct subsets of the SDUST dataset (Bearing and Gearbox) are thoroughly quantified in Table 9.

Table 9: Ablation results of the baseline CNNC and the proposed FDAN with or without FFT preprocessing across DIGR and SDUST Datasets. (%)

Model	FFT Preprocessing	DIGR Dataset	SDUST (Bearing)	SDUST (Gearbox)
CNNC	✗	49.12	58.22	28.50
CNNC	✓	52.62	75.53	77.26
FDAN	✗	53.12	85.26	28.88
FDAN	✓	62.86	92.28	97.34

As presented in Table 9, replacing the frequency-domain inputs with raw time-domain waveforms induces a universal performance degradation across all tasks. For the SDUST (Bearing) tasks, the performance drop is relatively moderate (CNNC drops from 75.53% to 58.22%; FDAN drops from 92.28% to 85.26%). This is primarily because the bearing experimental tasks are configured as variable-speed scenarios with overlapping speed distributions across domains, allowing the network to interpolate certain domain-shared temporal cues from adjacent states.

In stark contrast, for the 7-class SDUST (Gearbox) domain generalization tasks under isolated fixed speeds, a catastrophic performance collapse is observed when operating in the time domain (CNNC plummets to 28.50%; FDAN collapses to 28.88%). Under such rigid domain shifts, the substantial displacements of fault impact intervals in the time domain completely disrupt the convolutional filters. Because raw time-domain features lack spatial alignment, the convolutional backbone fails to extract any genuine, generalizable fault semantics. Consequently, due to the failure of the underlying architectural backbone, the high-level generalization capabilities of FDAN are also entirely bottlenecked, leading to a near-random classification performance (around 28.57%). Conversely, when FFT is enabled, the abnormal impact frequencies are explicitly unmasked and tracking of structural fault harmonics is smoothly restored, enabling FDAN to achieve an outstanding accuracy of 97.34%.

8 Additional Experimental Validation on HUST and CWRU Datasets:

To thoroughly evaluate the generalization capability of the proposed method, two publicly available bearing datasets are utilized in this study.

The first dataset is the HUST bearing dataset, which contains vibration signals collected from a Spectra-Quest Mechanical Fault Simulator. This dataset encompasses three distinct bearing health conditions: normal state, inner race fault, and outer race fault. All vibration signals were acquired at a sampling frequency of 25.6 kHz, with each record lasting for a temporal duration of 10.2 seconds. To simulate diverse operational conditions, data collected under ten varying rotational speeds (20, 25, 30, 35, 40, 60, 65, 70, 75, and 80 Hz) are employed for the cross-domain tasks.

The second dataset is provided by the Case Western Reserve University (CWRU) Bearing Data Center. In this work, the drive-end bearing vibration data sampled at 48 kHz are selected. The dataset comprises a normal health

state alongside two fault modes (inner race and outer race defects). Each fault mode includes three different defect severities (0.007, 0.014, and 0.021 inches), resulting in a total of seven distinct health conditions. These signals were collected under four slightly varying motor speeds: 1797, 1772, 1750, and 1730 rpm.

For both datasets, the raw vibration signals are partitioned into non-overlapping segments, with each segment containing 4096 data points. The domain generalization experimental configurations are summarized in Table 10.

Table 10: Description of DG Tasks on the HUST bearing (RT11–RT15, in Hz), and CWRU (T11–T15, in rpm) Datasets.

Task	Source Domains	Target Domains
RT11	20, 60	25, 30, 35, 40, 65, 70, 75, 80
RT12	20, 40, 60, 80	25, 30, 35, 65, 70, 75
RT13	20, 35, 40, 60, 75, 80	25, 30, 65, 70
RT14	20, 25, 35, 40, 60, 65, 75, 80	30, 70
RT15	20, 25, 30, 35, 40	60, 65, 70, 75, 80
RT16	1770, 1750, 1730	1797
RT17	1797, 1750, 1730	1770
RT18	1797, 1770, 1730	1750
RT19	1797, 1770, 1750	1730

For the HUST dataset, each rotational speed is treated as an independent domain. Under each specific speed condition, 60 samples are randomly selected to construct the training set, while another 120 samples are allocated for testing. The dataset is strictly balanced across all fault types to prevent class bias. During the training phase, the model is optimized for 100 epochs using a batch size of 16.

For the CWRU dataset, under each speed condition, only 7 samples are extracted for training—exactly one sample for the normal state and one for each of the six fault configurations. Meanwhile, 100 samples are assigned to the test set for evaluation. Under this extreme few-shot setting, the network is trained for 30 epochs with a batch size of 4.

Table 11: Diagnostic Accuracy of Different Methods Under (Tasks RT1 - RT6). (%)

Method	Task (HUST)					HUST Mean	Task (CWRU)				CWRU Mean
	RT11	RT12	RT13	RT14	RT15		RT16	RT17	RT18	RT19	
CNNC	77.16 ±2.47	81.55 ±3.01	78.58 ±6.10	80.50 ±2.60	71.33 ±6.92	77.82	88.00 ±3.08	90.20 ±2.38	87.80 ±1.30	86.00 ±6.74	88.00
CCDG	74.40 ±6.32	78.50 ±2.28	79.16 ±8.04	83.16 ±4.90	74.66 ±7.70	77.97	87.60 ±6.06	83.80 ±4.20	86.00 ±3.31	82.40 ±4.03	84.95
CCN	77.67 ±2.94	85.89 ±3.65	86.04 ±1.53	86.83 ±1.41	77.56 ±7.48	82.79	97.20 ±3.11	92.20 ±3.19	87.40 ±7.02	93.60 ±3.64	92.60
DGNIS	86.29 ±2.85	87.88 ±0.57	90.58 ±0.75	85.83 ±5.59	78.40 ±0.89	85.79	88.80 ±3.27	91.60 ±6.18	84.20 ±3.03	87.00 ±7.34	87.90
DDDG	86.37 ±6.09	86.16 ±6.16	82.41 ±5.63	82.83 ±3.31	77.13 ±5.64	82.74	98.20 ±2.68	94.80 ±3.96	91.00 ±2.23	94.60 ±3.91	94.65
CDDG	81.79 ±3.19	82.88 ±7.53	86.25 ±2.06	83.50 ±3.30	75.93 ±3.89	82.07	95.00 ±2.54	91.80 ±2.16	91.40 ±2.19	86.00 ±7.00	91.05
IEDGNet	87.56 ±1.86	89.32 ±1.67	90.01 ±1.93	89.70 ±1.47	80.06 ±1.57	87.33	93.20 ±4.65	91.60 ±2.30	87.20 ±3.03	86.60 ±2.88	89.65
FDAN	87.16 ±2.47	91.55 ±3.01	89.78 ±7.07	90.50 ±2.60	81.13 ±4.03	88.02	98.20 ±1.30	97.00 ±2.00	97.40 ±0.89	95.20 ±2.48	96.95

As shown in Table 11, FDAN consistently exhibits superior performance on these additional datasets. On the HUST dataset, it achieves performance comparable to the strong data augmentation method IEDGNet, showing notable improvements across all tasks. Crucially, on Task RT15, a highly challenging large-span generalization scenario from low to high speeds, FDAN demonstrates exceptional efficacy, outperforming the baseline by approximately 10%.

Furthermore, on the CWRU dataset, the extremely limited training data degrades the performance of typical data augmentation methods like IEDGNet. While causal inference-based (CCN) and disentanglement-based (DDDG)

methods perform relatively well in this few-shot setting, FDAN, leveraging its classification loss-driven augmentation strategy, successfully overcomes the limitations of small sample sizes. It achieves the strongest overall performance, outperforming the baseline CNNC by a significant margin of 8.95%. These comprehensive results further substantiate that FDAN’s performance gains are robust, versatile, and highly applicable to a wide variety of practical diagnostic scenarios.

9 Statistical Significance Analysis Across Multiple Datasets:

To mathematically evaluate the robustness of the proposed FDAN framework, we have conducted two levels of exploratory statistical testing: a dataset-level Paired t -test and a global-level Wilcoxon Signed-Rank Test.

Specifically, the Paired t -test assesses the significance of accuracy gains within individual datasets by evaluating paired performance differences on identical tasks. Concurrently, the Wilcoxon Signed-Rank Test evaluates the model’s global superiority across all datasets by analyzing the rank of these differences, which effectively neutralizes the extreme disparity in absolute baseline difficulties.

Firstly, to evaluate the performance stability on a per-dataset basis, we utilized the Paired t -test. As shown in Table 12, FDAN achieves statistically significant improvements ($p < 0.05$) against the vast majority of baseline models across different datasets. It is worth noting that against highly sophisticated SOTA models like IEDGNet or CDDG on specific extreme datasets (e.g., DIGR), the p -value occasionally exceeds 0.05. This is mathematically expected: in datasets with extreme domain gaps where the absolute performance ceiling is inherently constrained, the absolute margin of improvement is compressed, rendering the local p -value less sensitive.

Table 12: Paired t -test p -values matrix for FDAN against baseline models across individual datasets. (* indicates statistical significance with $p < 0.05$).

Dataset	CNNC	CCDG	CCN	DGNIS	DDD	CDDG	IEDGNet
DIGR	0.0031*	0.0076*	0.0128*	0.0107*	0.0046*	0.0521	0.1422
SDUST(bearing)	0.0002*	0.0008*	0.0008*	0.0043*	0.0304*	0.0714	0.1149
SDUST(gearbox)	0.0001*	0.0001*	0.0000*	0.0010*	0.0116*	0.0003*	0.0069*
HUST	0.0000*	0.0018*	0.0100*	0.0860	0.0160*	0.0024*	0.2199
CWRU	0.0012*	0.0003*	0.1251	0.0113*	0.2093	0.0179*	0.0102*

Secondly, evaluating the global superiority of a classifier across multiple datasets with vastly different fundamental difficulties requires a specialized non-parametric approach. Directly pooling raw accuracies for standard t -tests is statistically invalid due to the severe inflation of inter-task variance. Therefore, we adopted the standard for comparing classifiers over multiple datasets in machine learning [R8]: the Wilcoxon Signed-Rank Test. This test strictly evaluates the rank of paired performance differences independently of the absolute magnitude of the baseline accuracies.

Table 13: Global Wilcoxon Signed-Rank Test results across all evaluated tasks. Alternative hypothesis: the diagnostic accuracy of FDAN is significantly greater than the baseline model.

Baseline Model	W-Statistic	Overall p -value
CNNC	300.0	<0.0001
CCDG	300.0	<0.0001
CCN	300.0	<0.0001
DGNIS	299.0	<0.0001
DDD	276.0	<0.0001
CDDG	299.0	<0.0001
IEDGNet	290.0	<0.0001

As definitively demonstrated in Table 13, when comparing FDAN against all seven baseline models, the one-sided Wilcoxon test yields overall p -values of strictly < 0.0001 . This statistical evidence mathematically proves that the global performance improvements achieved by FDAN are highly deterministic and robust against random initialization variations across diverse industrial scenarios.